

Drone Agents 🤖: learning to fly to learn how to see

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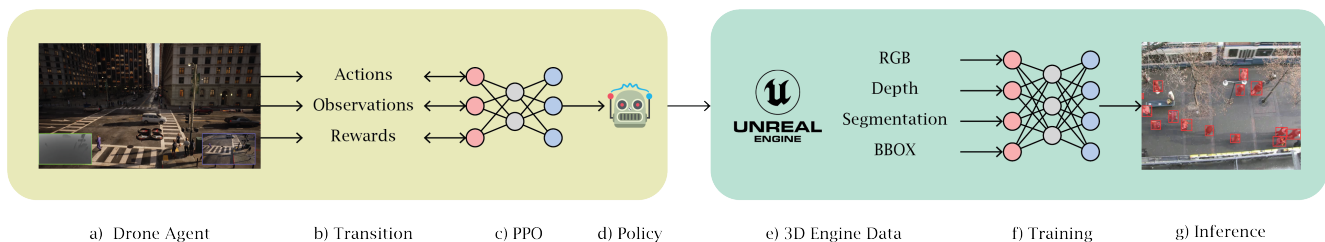


Figure 1: Overview of our method. We (a) design a realistic drone agent inside Unreal Engine 5 (UE5), with physically accurate movement capabilities, rgb, depth, and segmentation cameras. We then (b) define a set of possible actions, observations, and rewards for each agent in the environment, and (c) train the drone agents to complete a task (e.g., cover an area with high pedestrian density). After training, (d) the learned policy can be used by every agent to navigate the environment in real-time. (e) Inside UE5, each drone now can be used as an automatic data collector, that can create datasets in real-time. (f) The collected data can be used for multiple purposes, such as training or fine-tuning a pedestrian detector. (g) At inference time, we show how the collected data can be used to improve the generalization over both real and synthetic data.

Abstract

Recent advances in reinforcement learning (RL) have opened promising opportunities for autonomous drone navigation. However, bridging RL-based methods with high-fidelity simulation environments remains an open challenge. In this paper, we introduce a novel multi-task approach for real-time RL-based autonomous drone navigation inside Unreal Engine 5. The idea is simple but effective: by using a physically accurate drone model and systematically increasing the complexity of simulated scenarios — from predefined path following to dynamic visual tracking with advanced sensing modalities — we achieve impressive generalization to multiple dynamically changing tasks. Our method addresses critical gaps in drone autonomy research, such as obstacle avoidance in swarm coordination and robust visual target tracking under varying environmental conditions. Through comprehensive experimental evaluation on pedestrian detection data in an urban scenario, we highlight the essential factors influencing drone learning performance. Our fine-tuned YOLOv8

model improves pedestrian detection recall from 72.41% to 84.71% on synthetic aerial images, demonstrating the effectiveness of drone-collected data for domain adaptation. <https://mmlab-cv.github.io/DroneAgents/>

1. Introduction

The use of reinforcement learning (RL) for autonomous aerial navigation has garnered significant interest due to its potential to learn adaptable and robust control policies from high-dimensional sensory data. This paradigm shift is especially relevant for unmanned aerial vehicles (UAVs), which must operate in dynamic and uncertain environments where traditional model-based approaches—such as Model Predictive Control (MPC), waypoint tracking, and hand-crafted rule-based planners—struggle to generalize. Recent research has demonstrated the effectiveness of deep RL (DRL) in diverse UAV tasks including visual target tracking, obstacle avoidance, indoor navigation, and even champion-level drone racing. These DRL approaches out-

perform classical controllers through end-to-end learning and simulation-to-reality transfer. Nevertheless, integrating RL into photorealistic, physics-accurate simulations and scaling to real-world complexity remains a core challenge. In this work, we propose a unified multi-task RL framework for UAV navigation within the Unreal Engine 5 environment. To showcase the effectiveness of our method, we choose a coverage scenario in which drones in an urban environment are assigned the task of monitoring crowds of people. After being trained, the drone agents are used as automatic data collectors, that collect annotated images that we use to fine-tune a pedestrian detection model. Our results demonstrate that drone agents are capable of collecting high-quality data, which significantly enhances their ability to generalize to both unseen real-world and synthetic environments.

The main contributions of this work are as follows:

- We propose a multi-task reinforcement learning-based navigation pipeline for UAVs, developed within a photorealistic simulation environment in Unreal Engine 5 (UE5).
- We introduce a method for automated data collection using drone agents, addressing the current lack of aerial-view pedestrian datasets.
- We demonstrate improved generalization in pedestrian detection through a synthetic-to-real fine-tuning approach, leveraging the collected simulation data to enhance real-world performance.

Differently from other works in literature, our method runs completely in-engine with high quality fully synthetic data, unlocking new possibilities for photorealistic, faster than real-time UAV simulations that can generalize to real-world data after training.

2. Related Work

Recent advances in DRL for UAV navigation leverage methods such as Deep Q-Network (DQN), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO), enabling adaptive obstacle avoidance and robust path-following directly from visual inputs [5, 8, 14]. However, the integration of path following and obstacle avoidance remains a significant challenge, especially in scenarios involving limited field-of-view sensors such as low-cost LiDAR systems [12].

Target localization has benefited from cooperative multi-view approaches, addressing the ambiguity of monocular methods. Lin et al. introduced an end-to-end system using an IMM-UKF initialized via triangulation to fuse multi-view observations within a motion-adaptive nonlinear filter, achieving accurate real-world localization under onboard constraints [10].

Vision-based DRL has enabled GPS-free tracking and navigation. The VTD3 framework proposed by Zhao et al. integrates YOLOv8, BoT-SORT, and the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm to achieve stable visual tracking, significantly outperforming proportional-derivative (PD) controllers in dynamic environments [17]. Singla et al. proposed a memory-based DRQN with temporal attention and cGAN-based monocular depth estimation for energy-efficient indoor navigation under partial observability [15]. Wu et al.'s RELAX framework combines Hector-SLAM, RRT, and D3QN with 2D LiDAR, enabling lightweight UAV control in dynamic agricultural environments [16].

Sim-to-real transfer has been facilitated by platforms such as Unity ML-Agents [7], Unreal Engine's Learning Agents [3], and Neural Network Environments (NNE) [2], which support scalable training in physics-accurate, photorealistic simulations [1, 11, 4]. CAD2RL leverages domain-randomized simulation to train collision prediction from monocular RGB input, achieving strong generalization without real-world data [13].

Swift [9], a DRL-driven UAV system, demonstrated world-champion-level drone racing performance using on-board sensing, visual-inertial odometry, and data-driven residual models for perception and control—highlighting the effectiveness of hybrid learning in high-speed, agile tasks.

Building on these foundations, our approach integrates DRL with Unreal Engine 5 to train drone agents capable of autonomous navigation and data collection within high-fidelity simulated environments. This pipeline facilitates the fine-tuning of object detection models using the generated data, leading to enhanced performance across both synthetic and real-world domains.

3. Method

To implement our system, we adopt a reinforcement learning framework coupled with object detection, developed within the Unreal Engine 5 (UE5) environment. Utilizing UE5's Learning Agents toolkit, we extend the AirSim drone simulation to support diverse sensor modalities, such as RGB, depth, and segmentation cameras. The subsections that follow outline the training procedures used for both the autonomous drone agents and the pedestrian detection model.

3.1. Training the drones

Each drone agent is trained using a deep reinforcement learning algorithm (specifically, PPO—Proximal Policy Optimization) provided by the Unreal Engine Learning Agents toolkit. The training environment is designed with increasing complexity to ensure generalization, involving



Figure 2: Examples of our Drone Agents capturing setup. Each drone roams the city, following groups of pedestrians with a PTZ camera that can capture in multiple kinds of modalities (RGB, depth, segmentation, BBOX) and from multiple dynamically changing viewpoints.

static and dynamic pedestrian crowds and obstacles.

The reinforcement learning setup is formally described as a Markov Decision Process (MDP) defined by a tuple (S, A, R, T) , where:

- S is the set of states, represented by drone sensor observations (RGB images, depth images, segmentation masks, position, and orientation).
- A is the set of continuous actions: throttle, yaw, pitch, and roll.
- R represents the scalar reward, formulated to optimize drone behaviors.
- T represents state transition probabilities influenced by drone physics in UE5.

We define the reward function R_t at timestep t as a weighted sum of several components:

$$R_t = w_d R_d + w_a R_a + w_s R_s + w_o R_o + w_c R_c \quad (1)$$

where each term is defined as follows:

Maintaining fixed distance: Drones must maintain an optimal distance d_{opt} from the observed crowd:

$$R_d = -|||p_t - p_{crowd}||_2 - d_{opt}| \quad (2)$$

where p_t is the drone position and p_{crowd} is the centroid position of the observed crowd.

Camera alignment to the crowd: Ensures that the drone camera orientation aligns directly with the center of the crowd:

$$R_a = -\theta_t \quad (3)$$

where θ_t is the angular difference between the camera forward vector and the vector pointing from the drone to the crowd centroid.

Smooth movements: Penalizes abrupt changes in drone velocity, encouraging smooth trajectories:

$$R_s = -|v_t - v_{t-1}|_2 \quad (4)$$

where v_t is the drone velocity at timestep t .

Obstacle avoidance through rays: Avoidance of obstacles detected via raycasts emitted from the drone in multiple directions:

$$R_o = \sum_{i=1}^{N_{rays}} \begin{cases} -1, & \text{if ray}_i \text{ hits an obstacle within } d_{safe} \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where N_{rays} is the number of rays, and d_{safe} is the safe threshold distance.

Other drones avoidance: Penalizes proximity between drones to avoid collisions:

$$R_c = \sum_{j \in \text{drones}} \max(0, d_{safe, drone} - ||p_t - p_j||_2) \quad (6)$$

where p_j represents the position of drone j , and $d_{safe, drone}$ is the minimal acceptable inter-drone distance.

Weights w_d, w_a, w_s, w_o, w_c are hyperparameters tuned during training to balance task-specific requirements. For more insight on the details of the in-engine PPO algorithm, please refer to [11].

3.2. Training the detector

We employ YOLOv8 [6] for pedestrian detection, fine-tuning the model specifically for top-view imagery captured by drones—a particularly challenging viewpoint due to perspective distortion and frequent partial occlusions. The fine-tuning process uses a dataset automatically generated by our trained drone agents, and augmented with real-world images from COCO, containing approximately 33k annotated images (bounding boxes and segmentation masks).

Fine-tuning follows a standard transfer learning protocol:

1. Initialization from pre-trained YOLOv8 weights.
2. Training on the drone-collected top-view pedestrian dataset, optimizing a joint detection and segmentation objective:

$$\mathcal{L} = \mathcal{L}_{\text{box}} + \lambda_{\text{obj}} \mathcal{L}_{\text{obj}} + \lambda_{\text{seg}} \mathcal{L}_{\text{seg}} \quad (7)$$

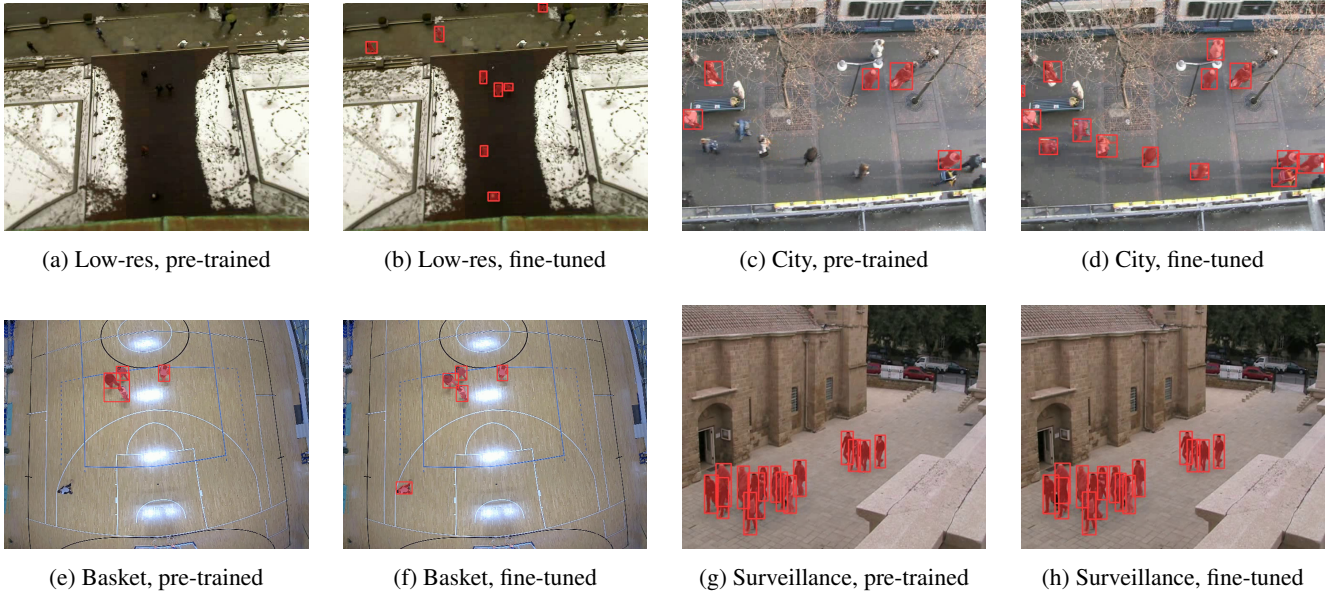


Figure 3: Qualitative comparison of YOLOv8 models. Left columns (a,c,e,g): Pre-trained baseline. Right columns (b,d,f,h): Fine-tuned model. Fine-tuning notably improves detection and segmentation of pedestrians from aerial views (low-resolution, crowded, and top-down perspectives).

where $\mathcal{L}_{\text{box}}$ is the bounding box regression loss, $\mathcal{L}_{\text{obj}}$ is the objectness (confidence) loss, and $\mathcal{L}_{\text{seg}}$ is the segmentation loss (e.g., pixel-wise cross-entropy). The weights λ_{obj} and λ_{seg} balance the contribution of each term.

3. Evaluation on both real (COCO) and synthetic (UE5-generated) datasets to ensure robust generalization.

This combined approach of reinforcement learning and supervised fine-tuning ensures that drone agents acquire robust navigation policies and generate high-quality data, ultimately improving pedestrian detection performance across diverse operational conditions.

4. Results

We conducted comprehensive experiments to evaluate the effectiveness of our multi-task RL-based drone navigation approach combined with YOLOv8 pedestrian detection. The results presented in Table 1 and Figure 4 demonstrate substantial improvements in generalization capability across our synthetic dataset. We observed an occasional imperceptible decrease in performance when testing the fine-tuned version on COCO, typical of the catastrophic forgetting associated to the naive finetuning. However, qualitative results in Fig. 3 clearly show that in both real and synthetic scenes, our method is able to better detect and segment under challenging conditions.

4.1. Quantitative Evaluation

Table 1 summarizes the quantitative results of YOLOv8, comparing the pre-trained baseline against our fine-tuned model using data collected by drone agents. On the synthetic test set, our fine-tuned detector significantly outperforms the pre-trained model across all metrics.

- A notable increase in bounding box recall, rising from 72.41% to 84.71%, indicating enhanced detection of pedestrians from challenging aerial viewpoints.
- Improved segmentation performance, with the $\text{AP}_{0.5}$ for masks increasing from 80.33% to 87.79%, demonstrating the effectiveness of our drone-collected dataset in training robust segmentation models.
- Higher precision across both detection and segmentation tasks, reflecting a reduction in false positives and greater reliability for real-time deployment scenarios.

On the real-world COCO dataset, performance remains comparable to the pre-trained baseline, indicating successful domain adaptation without compromising performance on traditional benchmarks. Notably, our fine-tuned model achieves slightly improved segmentation precision (51.85% $\text{AP}_{0.5-0.95}$) compared to the pre-trained baseline. Figure 4 provides precision-recall and F1 score curves for both models, illustrating that our fine-tuned model achieves consistently higher recall and precision values across both real and synthetic data.

		BBOX				MASK			
		$P(\%)$	$R(\%)$	$AP_{0.5}(\%)$	$AP_{0.5-0.95}(\%)$	$P(\%)$	$R(\%)$	$AP_{0.5}(\%)$	$AP_{0.5-0.95}(\%)$
TEST COCO	Pre-trained	84.23	74.90	84.80	64.01	84.04	73.18	82.74	51.52
	Fine-tuned	82.34	75.11	83.83	62.94	82.73	73.17	82.09	51.85
TEST SYNTH	Pre-trained	93.03	72.41	83.33	59.32	91.18	70.60	80.33	38.81
	Fine-tuned	93.17	84.71	91.92	67.05	91.23	82.05	87.79	44.94

Table 1: Performance of the two YOLOv8 models (pre-trained and fine-tuned) evaluated on detection (under the column “BBOX”) and segmentation tasks (under the column “MASK”), separately for the real domain (section “TEST COCO”) and synthetic domain (section “TEST SYNTH”).

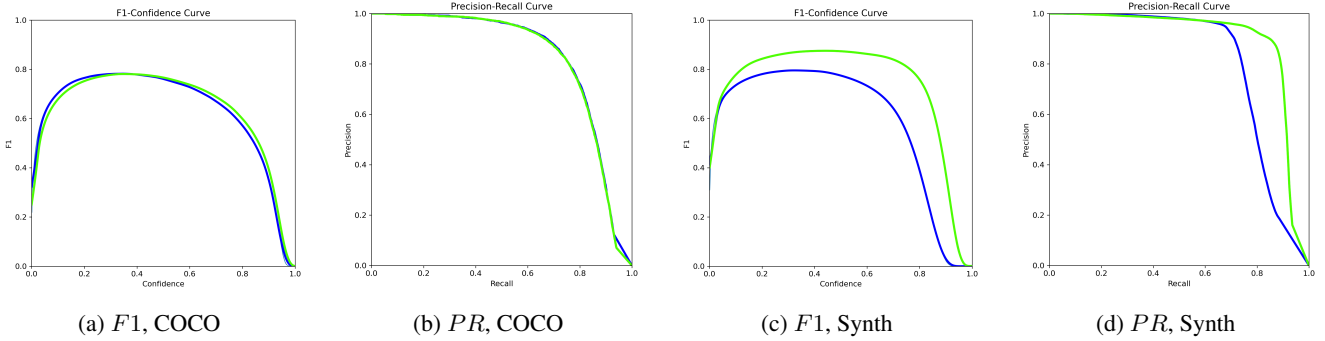


Figure 4: Precision-Recall (PR) and F1-score curves comparing YOLOv8 models before (blue) and after (green) fine-tuning. Left: COCO dataset. Right: Synthetic drone-collected dataset. The fine-tuned model consistently outperforms in both precision and recall.

4.2. Qualitative Evaluation

While the COCO dataset offers a robust baseline for general object detection and segmentation, it represents relatively simple scenarios characterized by standard perspectives, balanced lighting conditions, and well-separated objects. In contrast, datasets derived from trajectory prediction or sports-related contexts pose more significant challenges. These scenarios typically involve highly dynamic movements, significant occlusions, crowded scenes, unusual camera angles (such as the top-down view utilized by drones), and varying illumination conditions. Our fine-tuned model exhibits substantial improvements specifically in these challenging scenarios, highlighting the effectiveness of training with drone-collected data tailored to complex aerial perspectives and dynamic environmental conditions. The qualitative examples presented in Figure 3 visually illustrate the practical enhancements achieved through our approach:

- The fine-tuned model demonstrates superior performance on low-resolution images, successfully detecting pedestrians that are missed by the baseline (Figures 3a–b).
- In top-down urban scenes, the model accurately seg-

ments pedestrians under challenging conditions, including partial occlusions and suboptimal lighting (Figures 3c–d).

- In crowded scenarios, such as sports applications, the model exhibits improved discrimination of individuals within groups, leading to enhanced tracking performance (Figures 3e–f).
- In instances where the baseline YOLO model already performed well, our fine-tuned model maintains comparable performance (Figures 3g–h).

Overall, quantitative and qualitative evaluations confirm that our RL-trained drone agents can effectively collect diverse, high-quality training data. This data significantly enhances pedestrian detection models, resulting in improved robustness and generalization, critical for real-world drone-based monitoring and surveillance tasks.

5. Conclusions

In this study, we introduced a multi-task reinforcement learning framework for autonomous drone navigation integrated within Unreal Engine 5, coupled with a robust pedestrian detection pipeline using YOLOv8. Our approach suc-

cessfully bridges high-fidelity simulation with real-world applicability, addressing key challenges in drone autonomy including dynamic obstacle avoidance, crowd monitoring, and robust visual tracking from challenging viewpoints. Through comprehensive quantitative and qualitative evaluations, we demonstrated substantial gains in detection accuracy and generalization, particularly in scenarios characterized by perspective distortion, occlusion, and environmental complexity. Future research will focus on extending this framework to even more challenging environments and exploring real-world deployments.

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